

Lecture 18

ChE 467/596-012 Polymer Rheology

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of

In-class problem

Your synthetic chemist friend is crosslinking polymer A. At some point during the reaction, the material has passed the gel point. G' vs frequency data shows a nice flat line. Now, the theory of rubber elasticity says that $G' = nkT$ (n =number of crosslinks, k =Boltzmann constant and T =temperature in deg. K). To test this hypothesis, you measured G' of this sample at T_1 . You then heated the sample to T_2 where $T_2 = 2T_1$ (all temps are in deg, K). However, you found that G' at T_2 is more than twice the G' at T_1 . Explain your answer.

In-class problem

A photocrosslinkable system is being crosslinked in the rheometer by shining UV light. You are interested in monitoring the gelation and obtaining the gel point by running frequency sweeps at different time intervals. The problem you have is that the sample gets fully crosslinked in 20 mins, and running a reasonable frequency sweep takes 10 minutes. Are there any options available for you to monitor this crosslinking and obtain gel point?

Recap & Agenda

Last Class—Thixotropy

Today

- Examples with thixotropy and physical gels
- Multiphase systems—foams & emulsions
- Start on suspensions

of

Multiphase Systems

- *Air in Liquid* _____ *Foams*
- *Liquid in Liquid* _____ *Emulsion*
- *Solid in Liquid* _____ *Suspension & Filled systems*

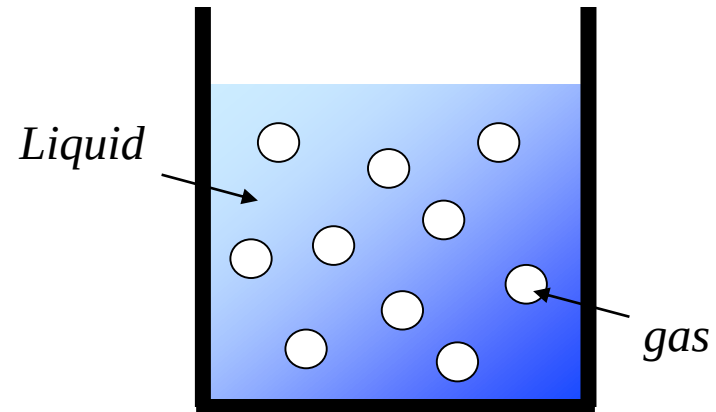
Foams and emulsions exhibit similar characteristics

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Foams and Emulsions

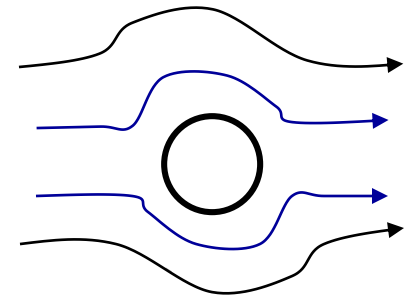
ϕ = gas volume fraction

When ϕ low $\Rightarrow$ isolated bubbles



Viscosity becomes higher/lower with bubble incorporation?

With bubbles, streamlines or flow distorted
 $\Rightarrow$ increased viscous energy dissipation



With low ϕ , surface tension is not important

of

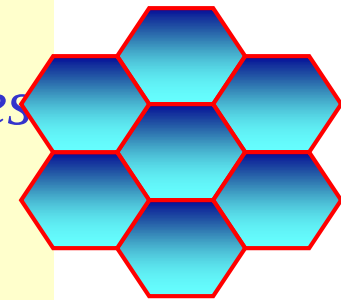
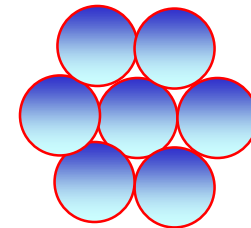
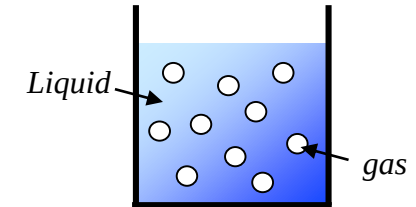
Foams and Emulsions, cont'd

High ϕ ($\phi > 0.64$):

- Surface tension begins to play a role
- Bubbles start to form close packs

As $\phi \rightarrow 1$:

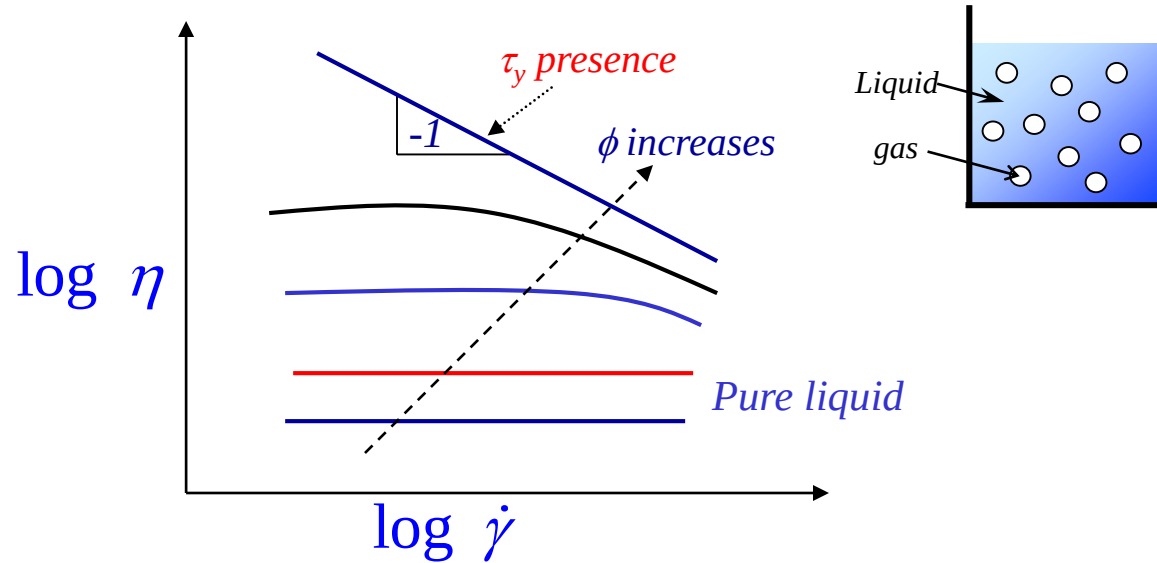
Bubbles form
2-dimensional-hexagonal like structures
3-dimensional-polyhedral structure
→ *dodecahedron (12 sided)*



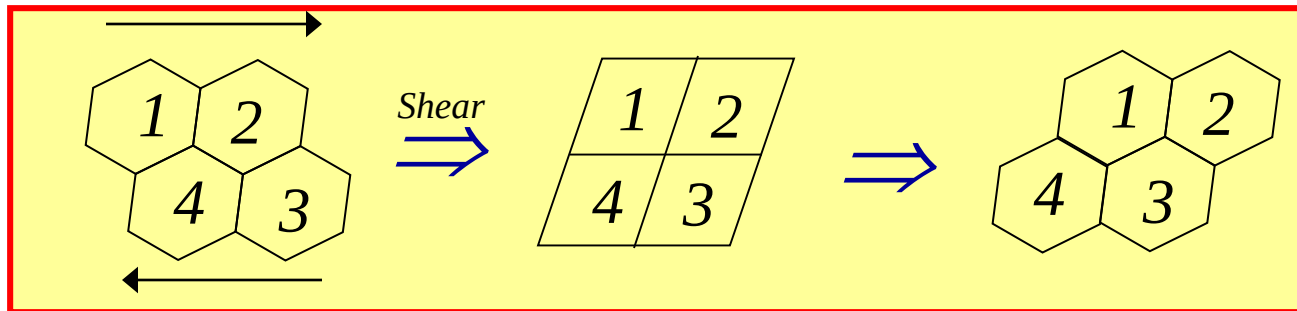
of

Foams and Emulsions, cont'd

Viscosity



At high $\phi \Rightarrow$ presence of τ_y

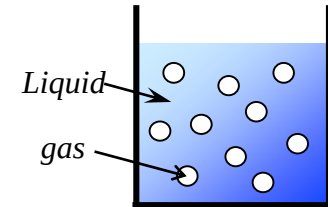
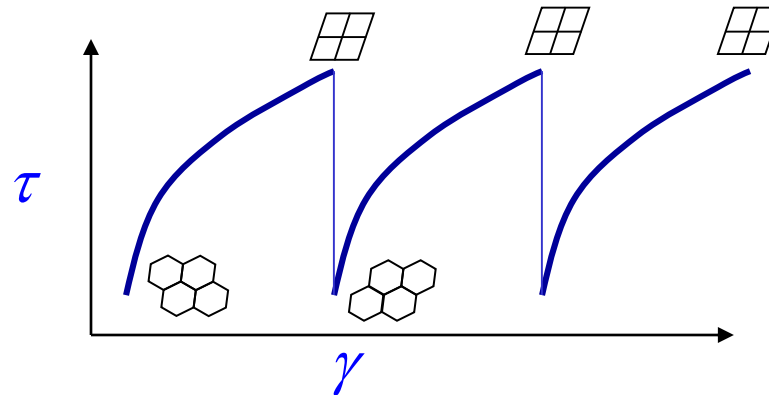


“Hopping” Behavior

of

Foams and Emulsions, cont'd

High ϕ



Surface tension

$$\tau_y = \frac{\sigma}{a} f(\phi) \quad \phi \geq 0.64$$

bubble size

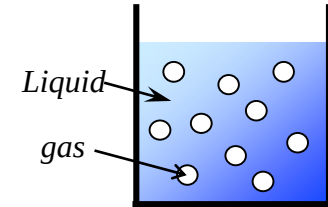
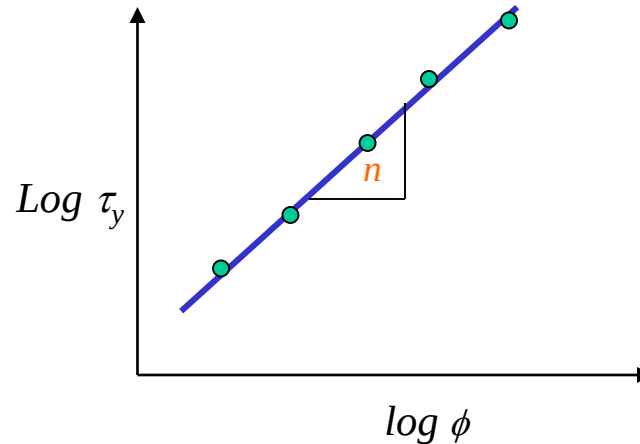
$$f(\phi) \sim \phi^n \quad \text{Experimentally observed}$$

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Foams and Emulsions, cont'd

High ϕ

$$f(\phi) \sim \phi^n$$



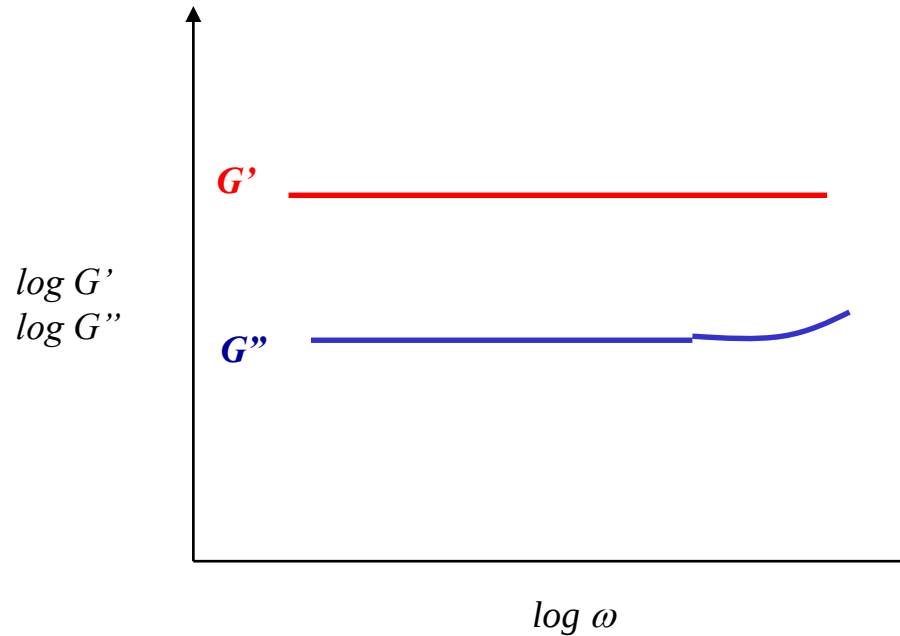
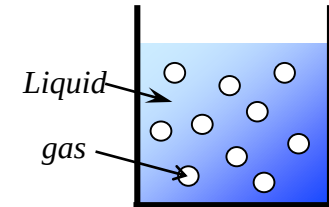
Assumptions:

- *Bubbles are “stable”*
- *Neglect coalescence*

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Foams and Emulsions, cont'd

High ϕ (G' , G'' vs ω)

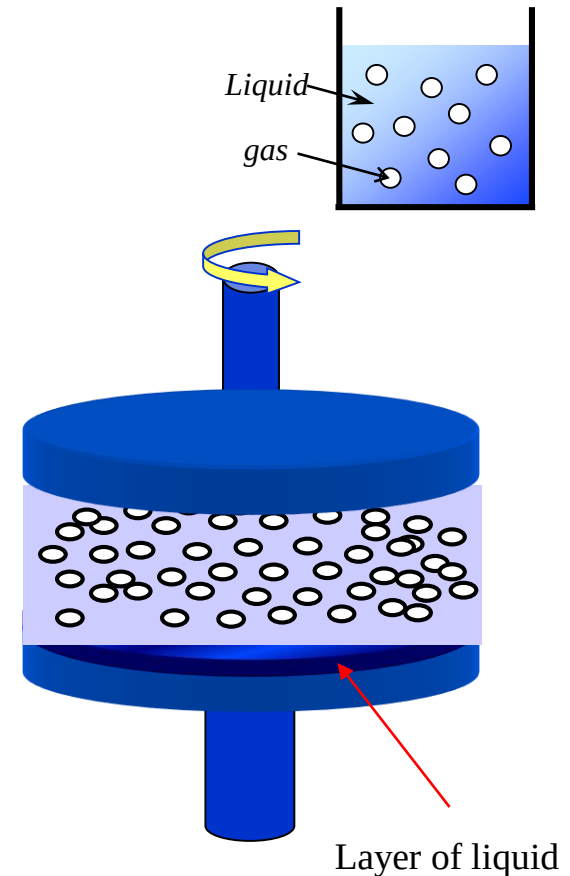


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Foams and Emulsions, cont'd

Wall slip occurs quite often

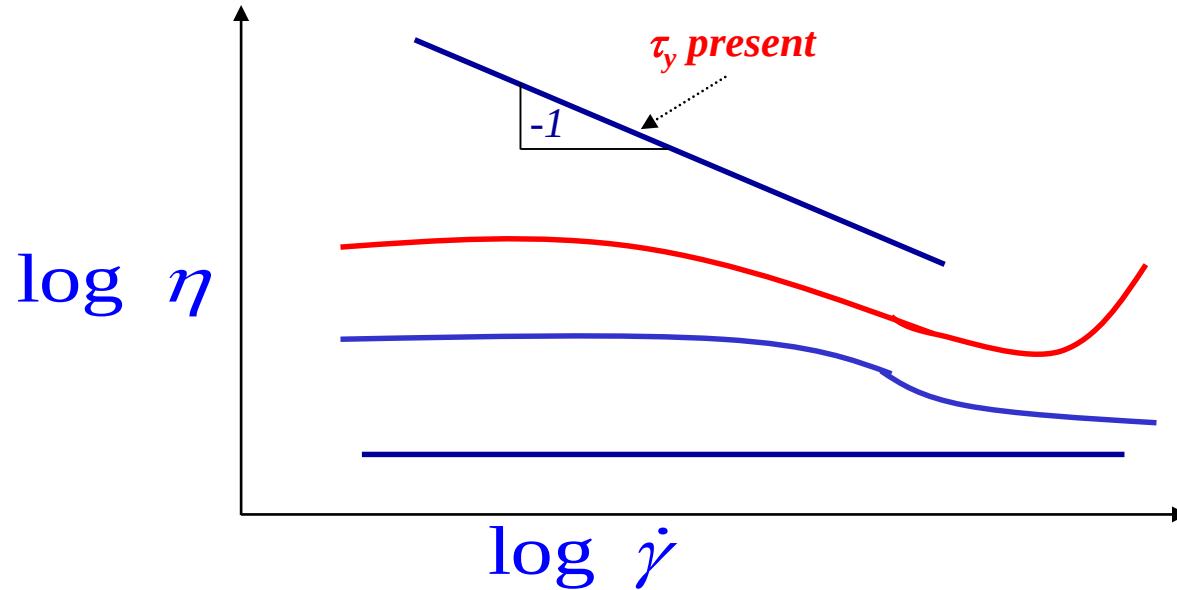
- Can use rough or serrated plates*
- Can use sand paper*
- Slip correction using measurements from 2 different gaps*



of

Suspensions

Spheres (solid) can generally show a range of steady shear behavior



of

Dilute Suspensions

Einstein (1906-1911)

$$\eta = \eta_{\text{solvent}} \left[1 + \frac{5}{2} \phi \right]$$

Assumptions:

1. **Creeping flow: no inertial effect*
2. *Surrounding fluid is Newtonian*
3. **No particle settling*
4. **No particle migration*
5. **No wall effect*
6. *Rigid spheres*
7. *No slip between particle and fluid*

of

Thixotropy/Gel Problem

You have created a fragile hydrogel that is easily broken when exposed to shear stress. However, when left undisturbed for a period of time after being broken, it can self-heal and regain its original gel structure. You want to further characterize this hydrogel.

- (a) (5) Describe what experiment(s) you could perform to calculate the maximum amount of shear stress the hydrogel can withstand before breaking.
- (b) (6) Describe what experiment(s) you could perform to calculate the time required for the hydrogel to self-heal after being broken.
- (c) (5) You notice that after exposing a new sample to very high shear stresses, the hydrogel rebuilds into an even stronger gel than it was before. Describe what might be happening to your sample at the microstructural level to explain this phenomenon.

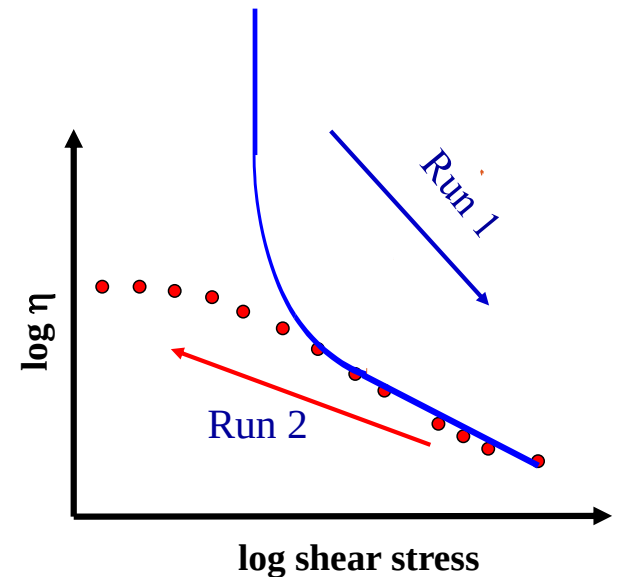
Summer 2017

Thixotropy/Gel Problem

of

Thixotropy/Gel Problem

Mr. X ran a viscosity-shear stress sweep in the rheometer of their novel complex systems, and found the data represented by a solid curve (Run 1). He then immediately did another experiment on the same sample in the rheometer to check for reproducibility but conducted a stress sweep in the reverse direction (shown as circles, Run 2) going from high to low shear stress.



- (a) Explain what type* of a fluid/material Run 1 data corresponds to.
- (b) Explain what type* of a material Run 2 reflects.
- (c) Explain why you think Runs 1 and 2 do not match. What is going on?

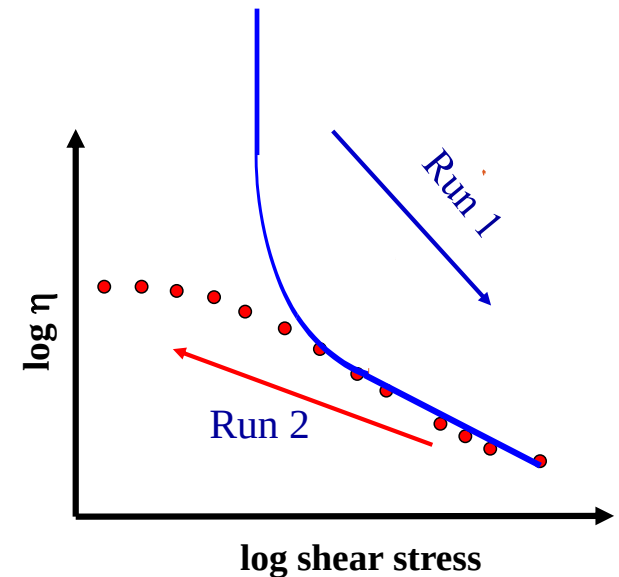
Spring 2017

Thixotropy/Gel Problem

of

Thixotropy/Gel Problem

(d) Mr. X did another experiment just as he did before with a fresh sample and found the same behavior as seen in Runs 1 and 2. So, now he did a third experiment, but instead of doing a stress sweep like Run 1, he measured the G' and G'' as a function of frequency. What would these curves look like qualitatively?



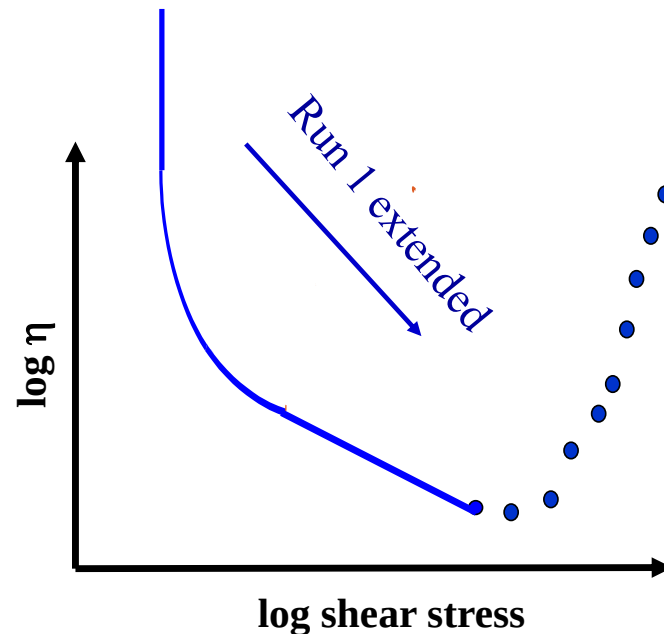
Spring 2017

Thixotropy/Gel Problem

of

Thixotropy/Gel Problem

(e) Finally Mr. X decided to perform another test (shown below). He decided to do Run 1 on another sample (same material) but this time went to a higher shear stress (data shown as circles). The graph he obtained has confounded him, in particular the behavior at high stress (circles). Please help him by explaining what is going on.



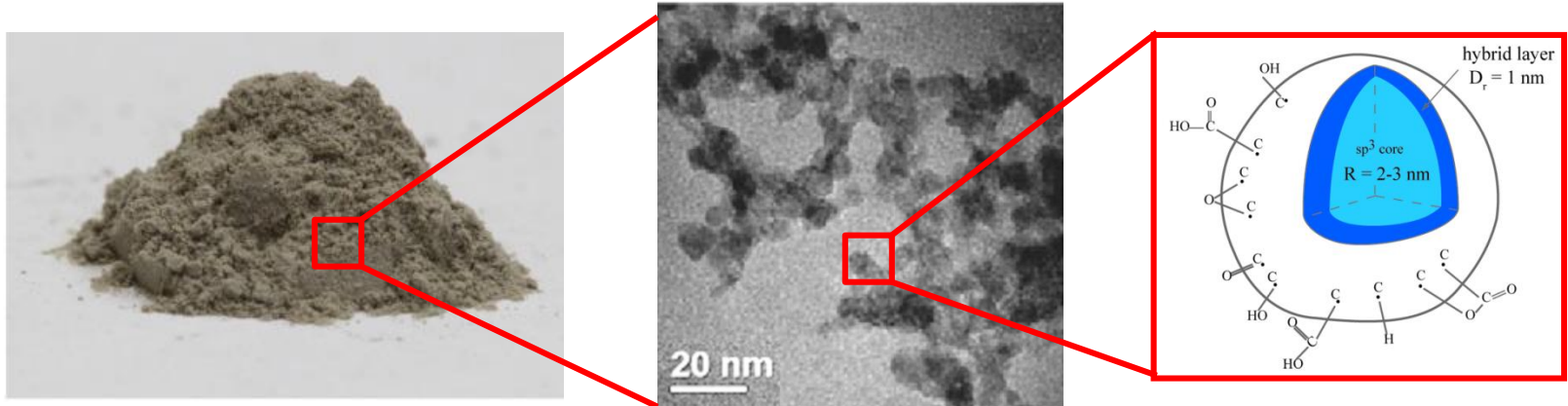
Spring 2017

Thixotropy/Gel Problem

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Nanodiamonds

- Developed in the former Soviet Union (1960s), made public in 1990s



- Core: sp^3 (bulk diamond properties)
- Shell: chemically active surface (highly functional)

Properties:

wear resistant
increased strength
thermally conductive
electrically resistive
thermally stable

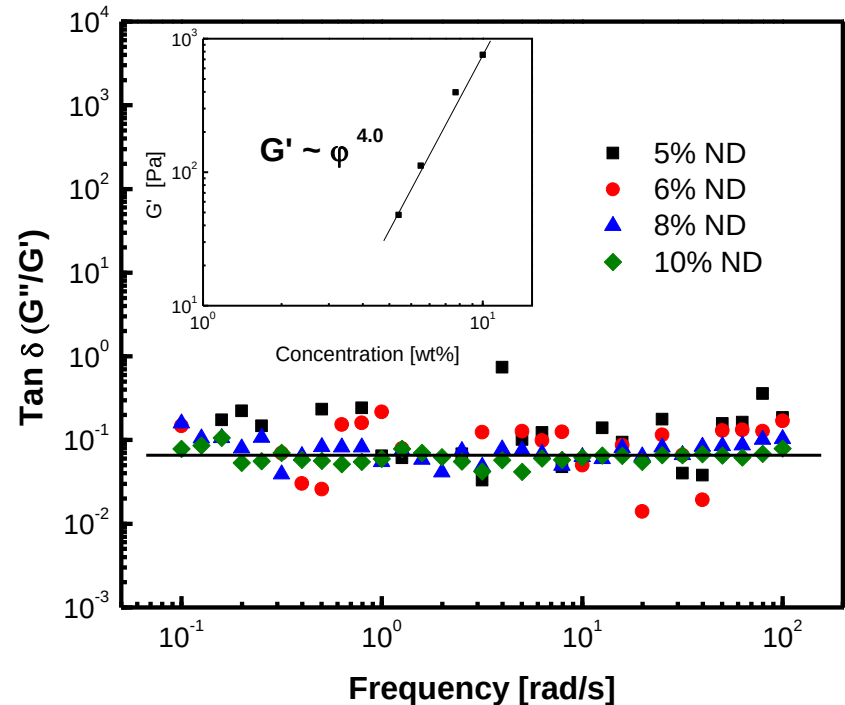
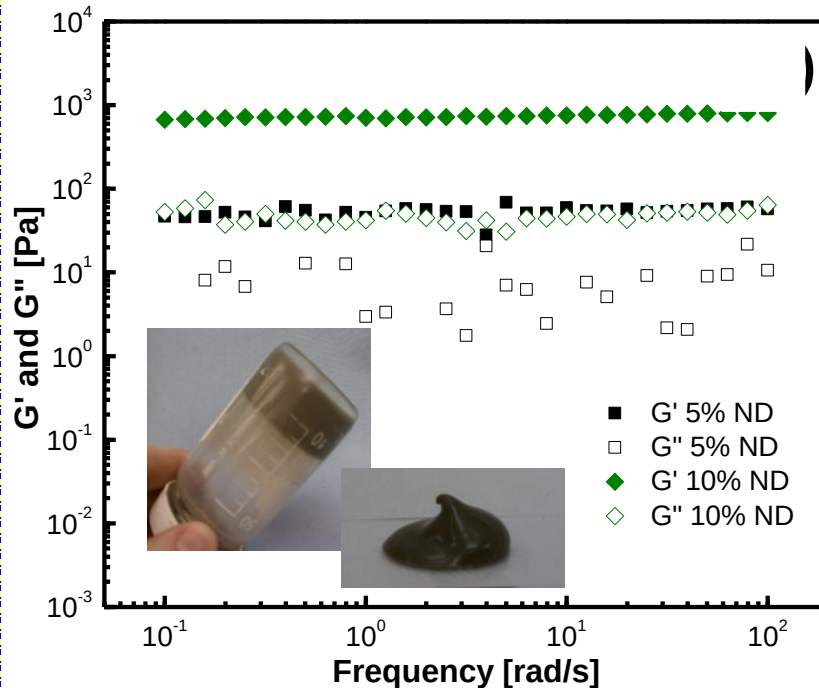
biocompatible
high surface area
narrow size distribution
chemically stable
antioxidant

absorbs proteins, enzymes, etc.
aids bone growth
penetrate cell walls
anti-carcinogenic
anti-inflammatory

of

Nanodiamonds in Mineral Oil

Instantaneous gel formation following mixing



G' follows a power-law behavior with ND concentration

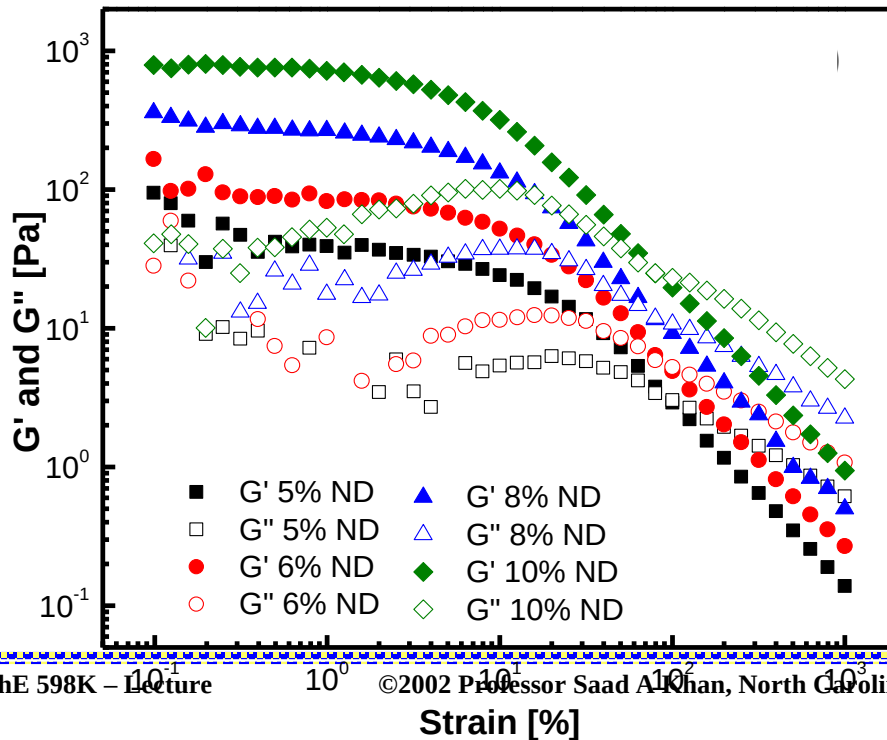
- Power-law exponent of 4 has been observed in other attractive colloidal systems
- Increasing G' with increasing ND content $\rightarrow$ aggregate bonds increase with ND content
- Equivalent increase in G''
- Loss tangent independent of ND concentration

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Structural Breakdown

Experiment: apply a strain sweep (0.1 -1000%)

- G' dominant for all concentration
- Structural breakdown observed with increasing strain
- G'' ultimately dominating
- Critical strain seems to be shifting to lower values with increasing ND concentration



Can we use this information to obtain useful material properties?

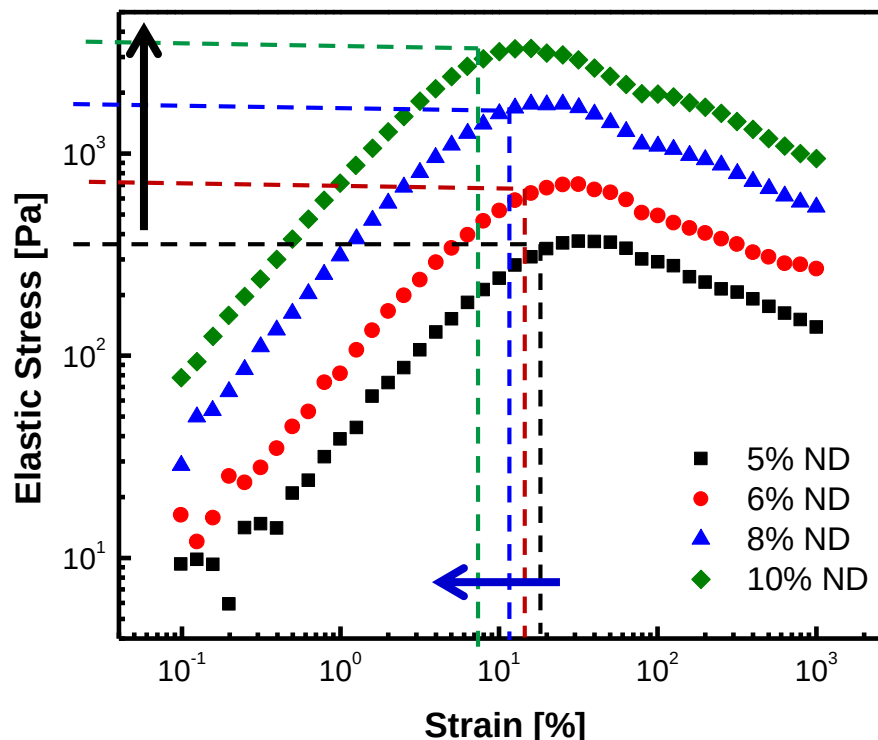
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Strain and Yield Stress

Yield stress measurement :

- Various approaches available (steady shear, creep, etc)
- Our preference-exploit dynamic data (Walls 2003, Macosko 1986)

Plot elastic stress ($G' \gamma$) as a function of strain



Yield stress (σ_y): maximum in elastic stress ($G' \gamma$)

- Increases with concentration

Critical Strain (γ_y): corresponding strain at the maximum

- Decreases with concentration

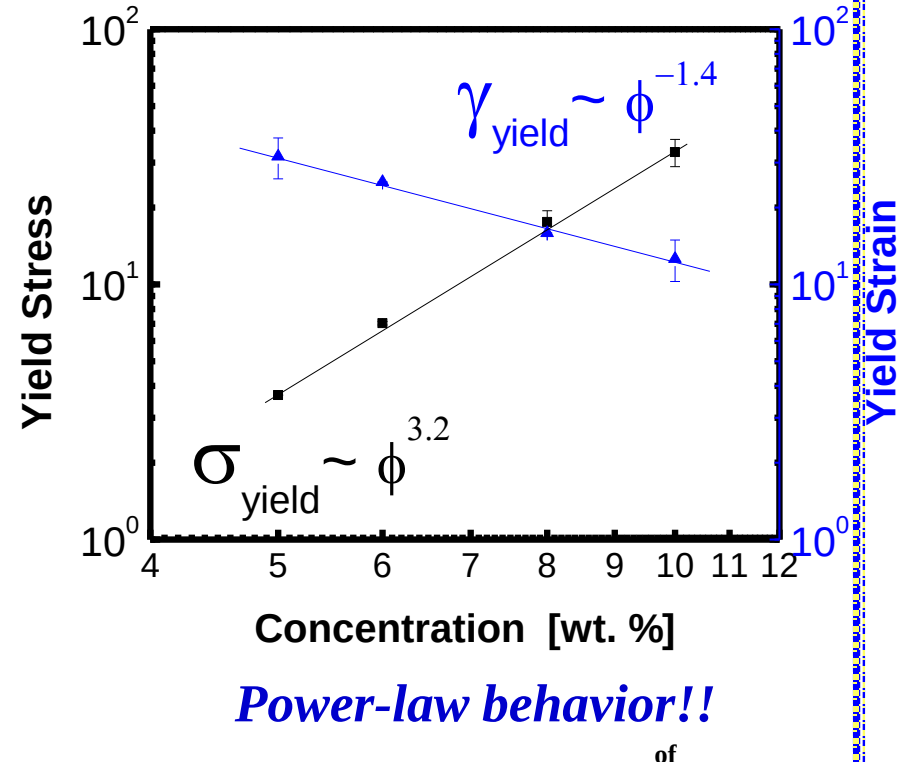
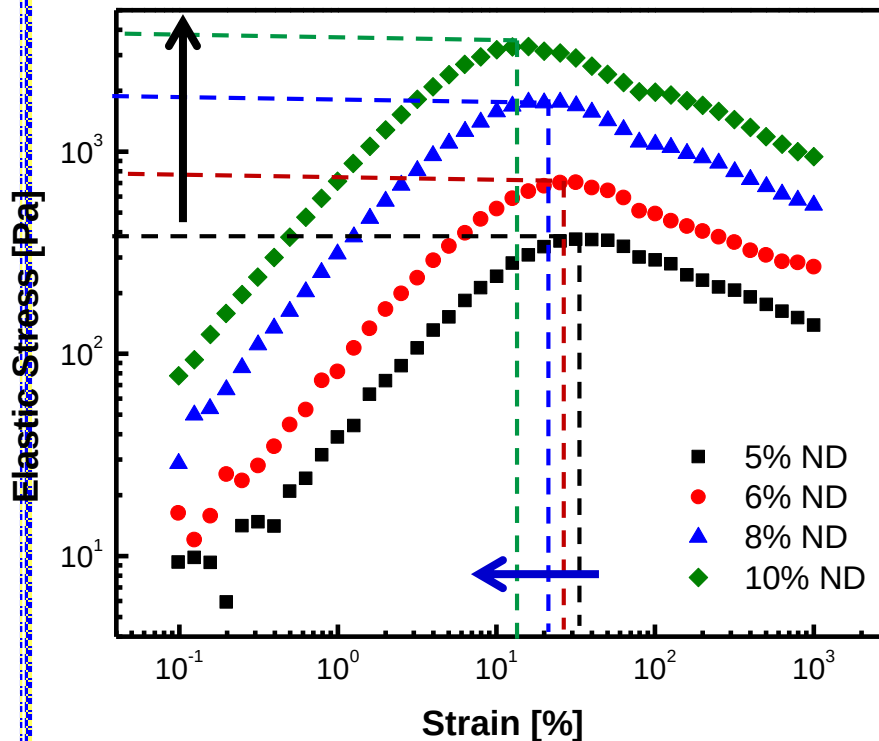
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Strain and Yield Stress

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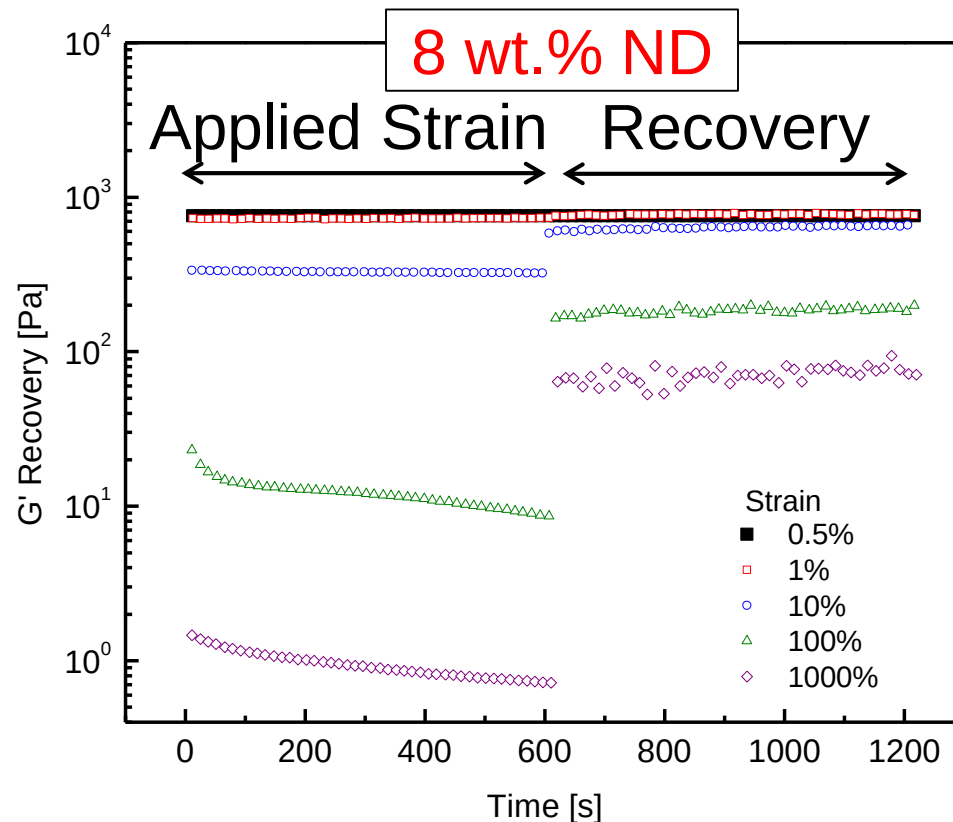
Plot elastic stress ($G' \gamma$) as a function of strain



Structural Breakdown and Recovery

Experiment:

- Apply a strain for 10 minutes, then monitor recovery
- Repeat for various strains



Breakdown:

- Strain dependent
- Majority occurs the first few cycles

Recovery:

- Instantaneous
- Only partial recovery!!!
- Varies with strain

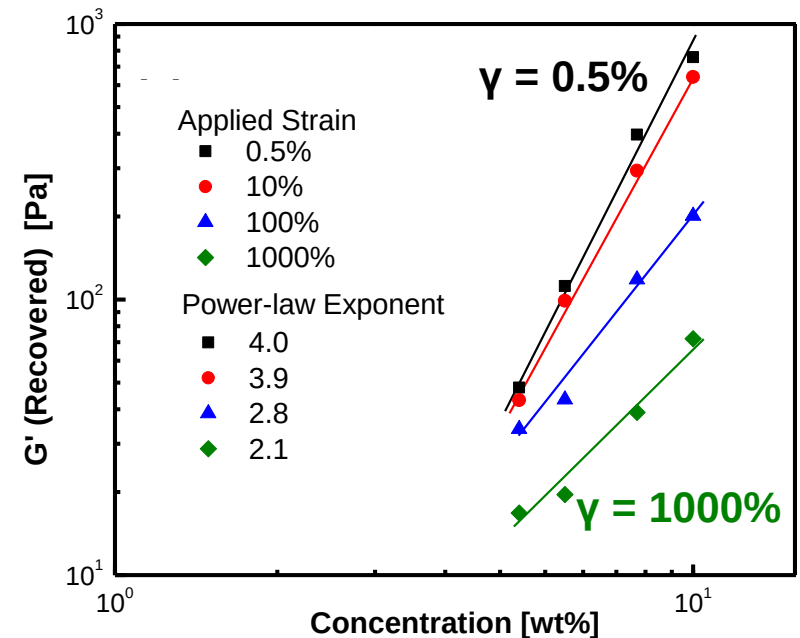
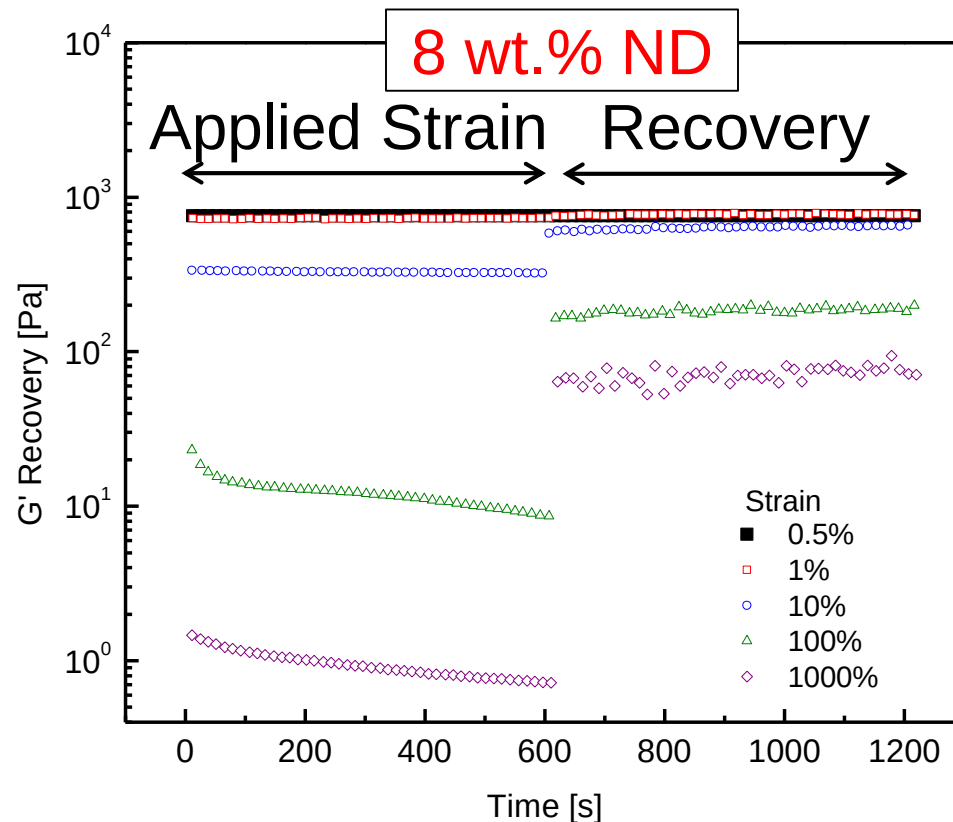
How is the recovery dependent on ND concentration?

Can we recover the original structure?

Structural Breakdown and Recovery

Experiment:

- Apply a strain for 10 minutes, then monitor recovery
- Repeat for various strains



Power-law behavior

- Suggests the mechanism of breakdown & recovery are independent of concentration

Different exponents

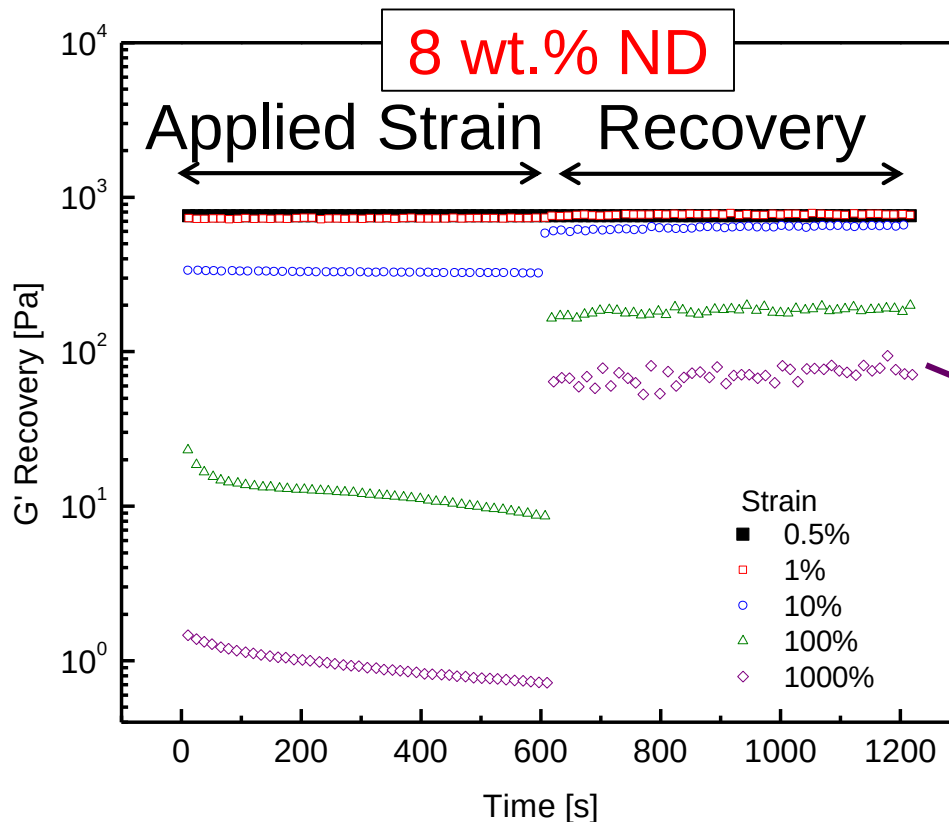
- Exponents vary with applied strain
- Different strains = different microstructures

Is the recovery dependent on ND concentration?

Structural Breakdown and Recovery

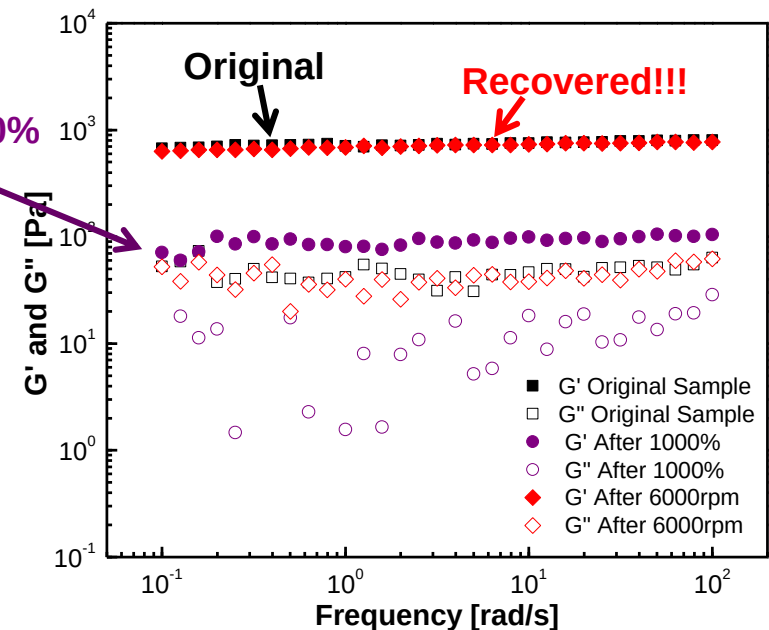
Experiment:

- Apply a strain for 10 minutes, then monitor recovery
- Repeat for various strains



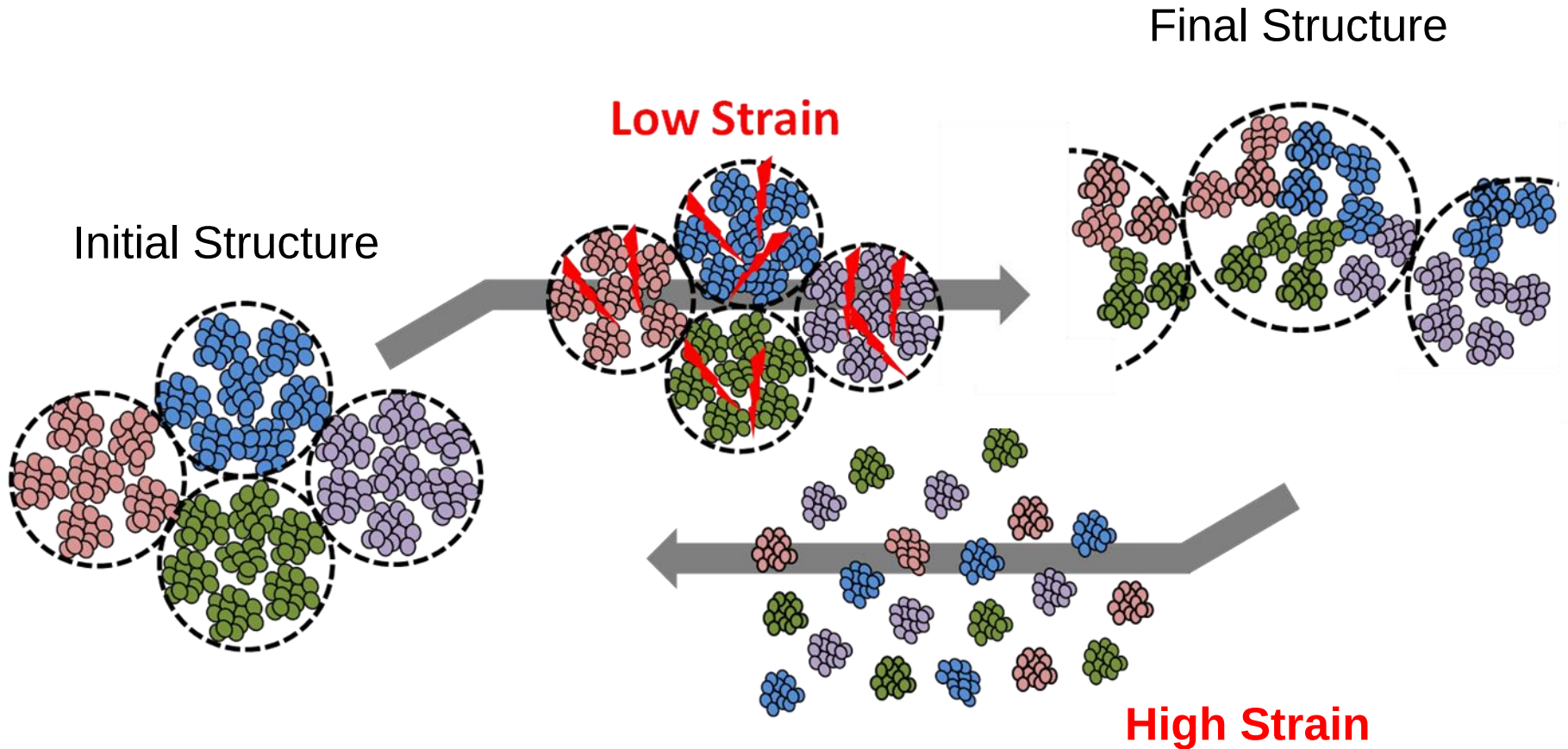
Can we recover the original structure?

High shear recovers the original microstructure instantaneously



What does this tell us about the material?

Structural Recovery After Strain



What interactions are occurring between the aggregates?

Dilute Suspensions, cont'd

$$\eta = \eta_{\text{solvent}} \left[1 + \frac{5}{2} \phi \right]$$

Assumption 1:

Creeping flow: no inertial effects

Re[#] has to be small (<<0.1)

If Re[#] is not small

$$\eta = \eta_s \left[1 + \frac{5}{2} \phi + 1.34 \text{Re}_p^{3/2} \right]$$

$$\text{Re}_p = \frac{\rho \dot{\gamma} a^2}{\eta_s}$$

Assumption 5:

No Wall effect

Gap has to be >> 10 particle diameter

of

Dilute Suspensions, cont'd

$$\eta = \eta_{\text{solvent}} \left[1 + \frac{5}{2} \phi \right]$$

Assumption 3:

No particle settling

$$\rho_s \cong \rho_p;$$

If $\rho_s \neq \rho_p$

Need to determine how long particles take to settle

$$t_{0.1h} = \frac{0.45\eta_s h}{\left| (\rho_p - \rho_s) \right| a^2 g}$$

a = particle radius

h = gap or height (Couette)

ρ_s = solvent viscosity

$t_{0.1h} \Rightarrow$ time required to settle 10% of rheometer gap or Couette height

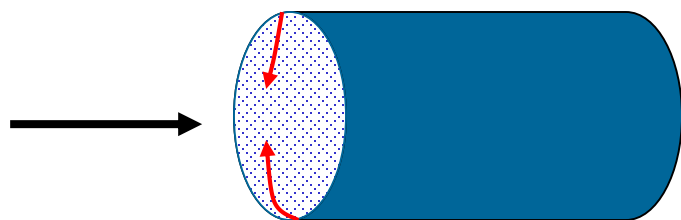
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Dilute Suspensions- cont'd

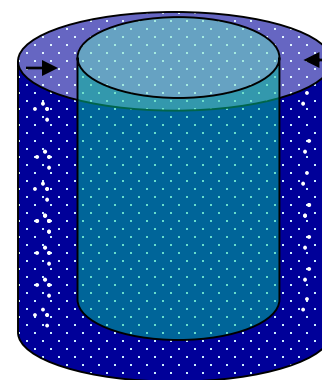
$$\eta = \eta_{\text{solvent}} \left[1 + \frac{5}{2} \phi \right]$$

Assumption 4:

No particle migration



$$K = \frac{\rho_p \bar{v}^2 a^4}{h k T}$$



$\bar{v} = \frac{1}{2} v_{\text{max}}$ in Couette

$\bar{v} = \frac{2}{3} v_{\text{max}}$ in Poiseuille flow

k Boltzman constant

T Temperature

If $K < 0.1$ Couette

$K < 0.01$ Poiseuille

Then we can neglect migration of particles

of

Generalized Einstein's Equation

Batchelor & Schowalter

$$\underline{\underline{\tau}} = \underline{\underline{\tau}}_s + \underline{\underline{\tau}}_p$$

*Stress contribution
from solvent*
*Stress contribution
from particles*

$$\underline{\underline{\tau}}_p = \frac{1}{V} \int \text{around the particle}$$

$$\underline{\underline{\tau}} = 2\eta_s \left[1 + \frac{5}{2} \phi \right] \underline{\underline{D}}; \quad 2\underline{\underline{D}} \equiv \text{Rate of strain tensor}$$

$$\eta = \eta_{\text{solvent}} \left[1 + \frac{5}{2} \phi \right]$$

Einstein's Equation

As ϕ increase, particle start to interact with each other

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Generalized Einstein's Equation, cont'd

$$\eta = \eta_{\text{solvent}} \left[1 + \frac{5}{2} \phi \right]$$



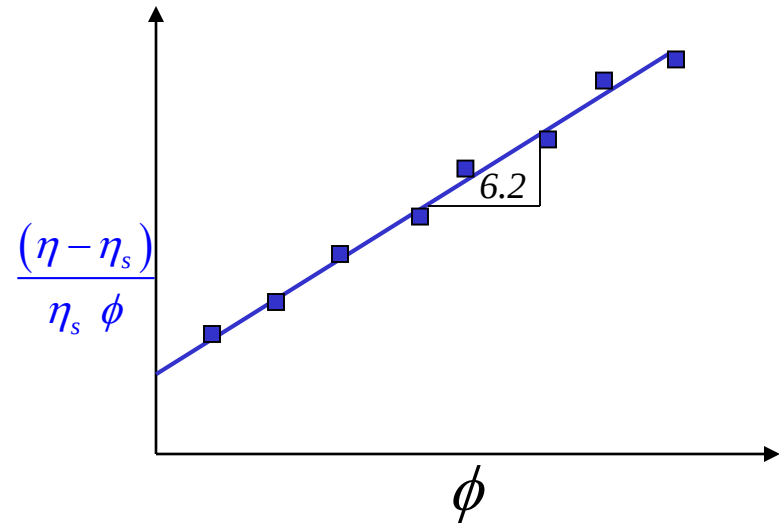
Takes into account interaction between particles

$$\eta = \eta_s \left[1 + \frac{5}{2} \phi + k\phi^2 + \dots \right]$$

Rearrange



$$\frac{(\eta - \eta_s)}{\eta_s \phi} = \frac{5}{2} + k\phi$$

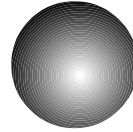


Experimentally $k=6.2$

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Dilute Suspensions - cont'd

Non interacting spheres:



$$\eta = \eta_{\text{solvent viscosity}} \left[1 + \frac{5}{2} \phi \right] \quad \text{Einstein Equation}$$

Volume fraction of filler

Deviation:

Spheres start interacting

$$\eta = \eta \left(1 + \frac{5}{2} \phi + 6.2 \phi^2 \right)$$

Non-submicron spheres => no colloidal forces present
As the sphere size $\leq 1 \mu\text{m}$, start having interaction between particles

of

Definitions

Relative or reduced viscosity:

$$\eta_r = \frac{\eta}{\eta_s}$$

Specific Viscosity:

$$\eta_{\text{specific}} = \frac{\eta - \eta_s}{\eta_s}$$

Intrinsic Viscosity:

$$[\eta] = \lim_{\phi \rightarrow 0} \frac{\eta_{\text{specific}}}{\phi}$$

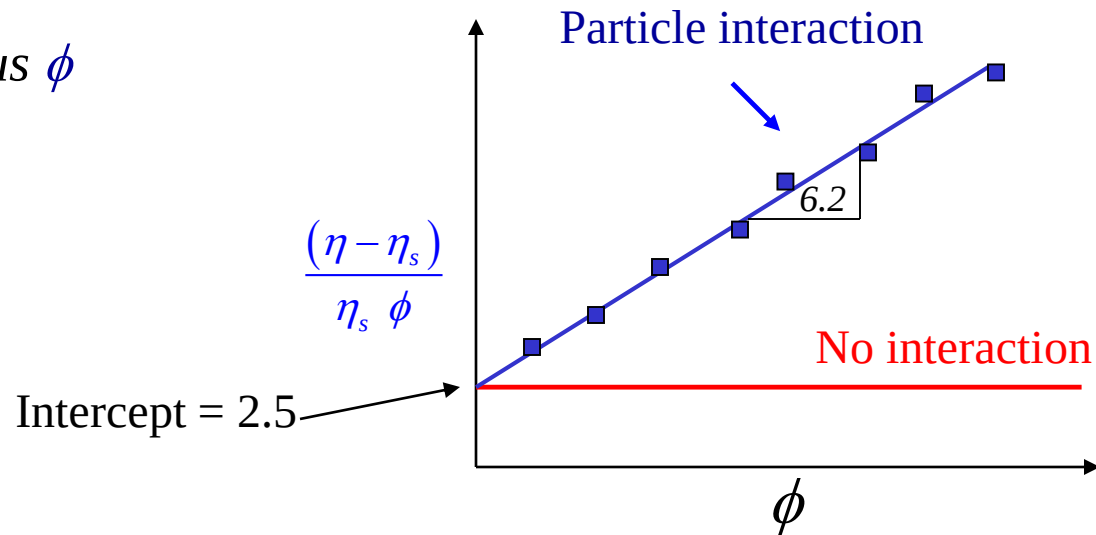
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In-class Problem

Problem: Given samples of different volume fraction of 2micron diameter glass beads in oil. Solvent & sample viscosities known for different ϕ .
How would you determine graphically if interactions between particles exist? What is the intrinsic viscosity of these suspensions?

Plot $\frac{(\eta - \eta_s)}{\eta_s \phi}$ versus ϕ

$$\frac{(\eta - \eta_s)}{\eta_s \phi} = \frac{5}{2} + k\phi$$



$[\eta] = 2.5$ for spherical suspensions (From Einstein Equation)

In-class Problem

Problem: Given PS latex (1 micron diameter) in water ($\eta = 1$ cp); $\phi = 0.05$;
 $\rho_p = 1.05 \rho_s$.

- (a) Obtain viscosity of suspension
- (b) Determine if inertia, migration or settling is important in couette system. $L = 50$ mm, $R_i = 25$ mm; $R_o = 26$ mm; $\Omega < 100$ rad/s

$$\dot{\gamma} = \frac{\Omega R_i}{R_o - R_i} \text{ in couette}$$

$$\begin{aligned} (a) \quad \eta &= \eta_{\text{solvent}} \left[1 + \frac{5}{2} \phi \right] \\ &= 1.13 \text{ cp} \end{aligned}$$

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In-class Problem

Problem: Given PS latex in water ($\eta = 1$ cp); $\phi = 0.05$; $\rho_p = 1.05 \rho_s$.

- (a) Obtain viscosity of suspension
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$$\text{Re}_p = \frac{\rho_p \dot{\gamma} a^2}{\eta_s} = 6.6 \times 10^{-6} \Omega$$

$$\dot{\gamma} = \frac{\Omega R_i}{R_o - R_i}$$

So, inertia not a problem

$$t_{0.1h} = \frac{0.45 \eta_s h}{|(\rho_p - \rho_s)| a^2 g} = 1.8 \times 10^5 \text{ sec}$$

Settling not a problem

of

In-class Problem

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- (a) Obtain viscosity of suspension
- (b) Determine if inertia, migration or settling is important in couette system. $L = 50$ mm, $R_i = 25$ mm; $R_o = 26$ mm; $\Omega < 100$ rad/s

$$\dot{\gamma} = \frac{\Omega R_i}{R_o - R_i}$$

$$K = \frac{\rho_p \overline{v^2} a^4}{h k T} = 0.0025 \Omega^2$$

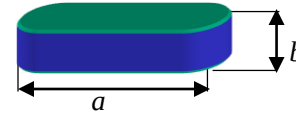
Migration negligible for $\Omega < 20$

$h =$ length of couette
avg. vel. $= \Omega R_i$

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Fibers - Dilute

Need to consider time scale of orientation/rotation of fiber



Aspect ratio, $r_p = \frac{a}{b}$

*Time scale of rotation / Brownian motion = $1/D_r$
(D_r is the rotary Brownian diffusion coefficient)*

$$D_r \cong \frac{kT}{8\pi\eta_s a^3}$$

For Spheres

$$D_r \cong \frac{3kT(\ln r_p - \frac{1}{2})}{8\pi\eta_s a^3}$$

For Prolate sphere

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Fibers-cont'd

Time scale of motion from flow $\sim \frac{1}{\dot{\gamma}}$

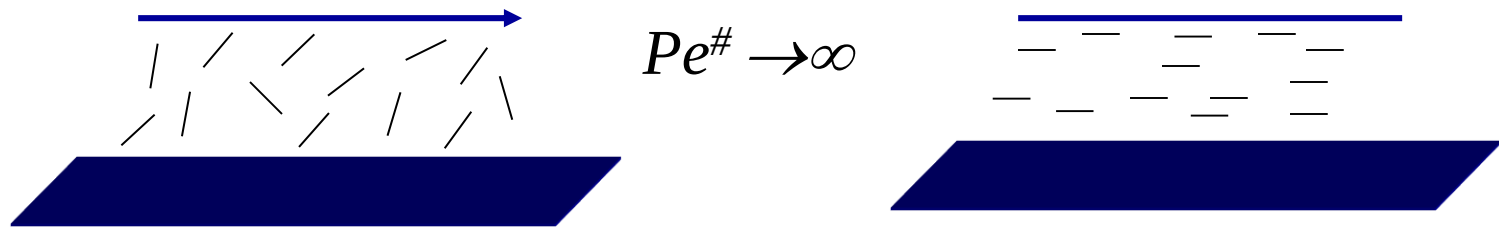
Define Peclet #

$$Pe^{\#} = \frac{\dot{\gamma}}{D_R}$$

$Pe^{\#} \rightarrow 0$ *Brownian motion dominates*

$Pe^{\#} \rightarrow \infty$ *Convective motion dominates*

As $Pe^{\#} \rightarrow \infty$, fibers orient along the flow direction

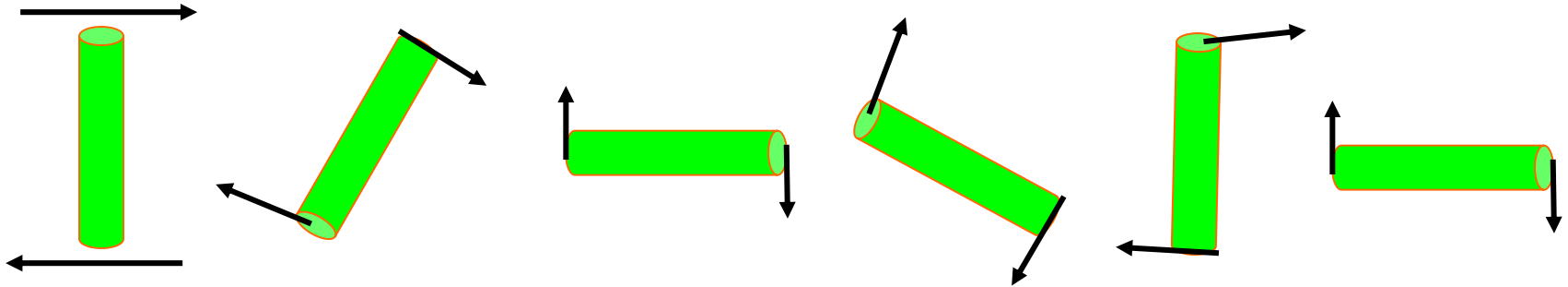


Reality: Fibers do orient but also rotate very slowly

of

Fibers-cont'd

Fibers do orient but also rotate/tumble very slowly



Slow Rotation: Time period for 1 oscillation

$$T_{period} = \frac{2\pi}{\dot{\gamma}} \left(r_p + \frac{1}{r_p} \right)$$

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Fibers-cont'd

Orientation:

$$\tan \phi = r_p \tan \left(\frac{\dot{\gamma} t}{r_p + \frac{1}{r_p}} \right)$$

$\phi = 0$



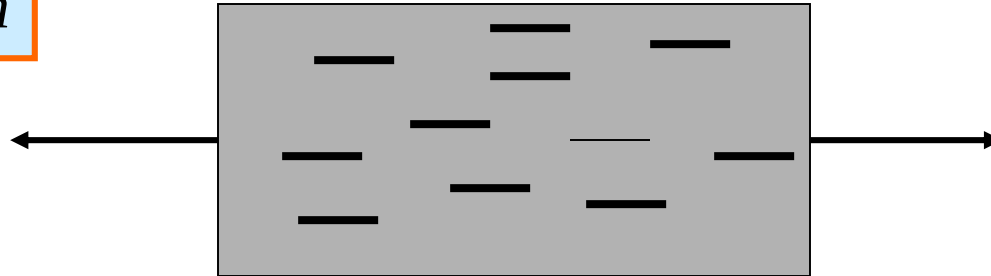
$\phi = 90$



of

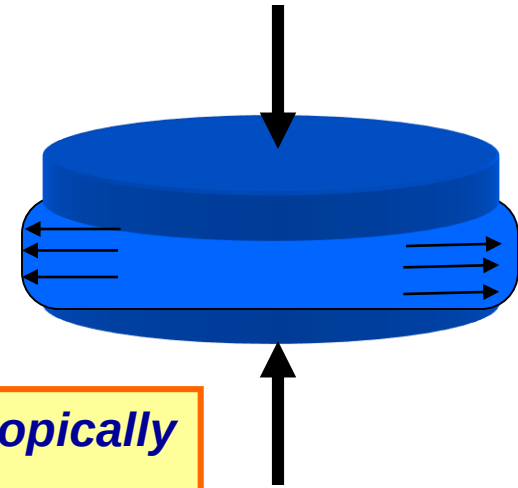
Fibers-cont'd

Extension



Ex: Squeezing flow between plates

Viscosity may be lower with fillers than without



- *you are applying extensional flow macroscopically*
- *have shear+extension at the micro level*

of

Recap & Approach

A. Dilute Systems

1. spheres (Einstein eqn.,)

2. fibers (brief)

concept of rotation

concept of $Pe^{\#}$

B. Non-dilute Systems

1. Interparticle forces & potential energy

2. Interpret rheology

of

Non-dilute Suspensions

~1% can show complex non-dilute behavior

Relative motion of neighboring particles

→ hydrodynamic interaction forces

=> refers to interactions caused by flow

(present in all systems)

Lot of times, deal with small fillers ($< 1 \mu\text{m}$)

Small fillers => additional interactions

of

Approach

B. Non-dilute Systems

1. Interparticle forces & potential energy

- **Attractive forces (dispersion)**
 - **van der Waals**
induced dipoles, possible in all systems
- **Repulsive forces**
 - **electrostatic**
 - **steric**

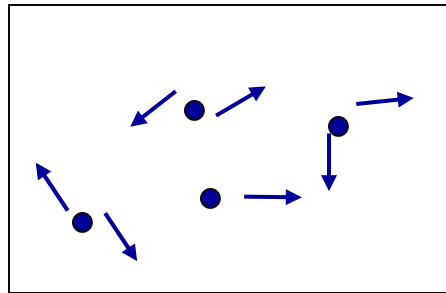
2. Interpret rheology (in terms of)

- **Hard spheres (Brownian)**
- **Stable system**
- **Flocculated system**

of

Non-dilute Suspensions-Brownian Motion

- *Tends to randomize things*
- *Concerned with translational diffusion coefficient*



of

Non-dilute Suspensions-Interparticle forces

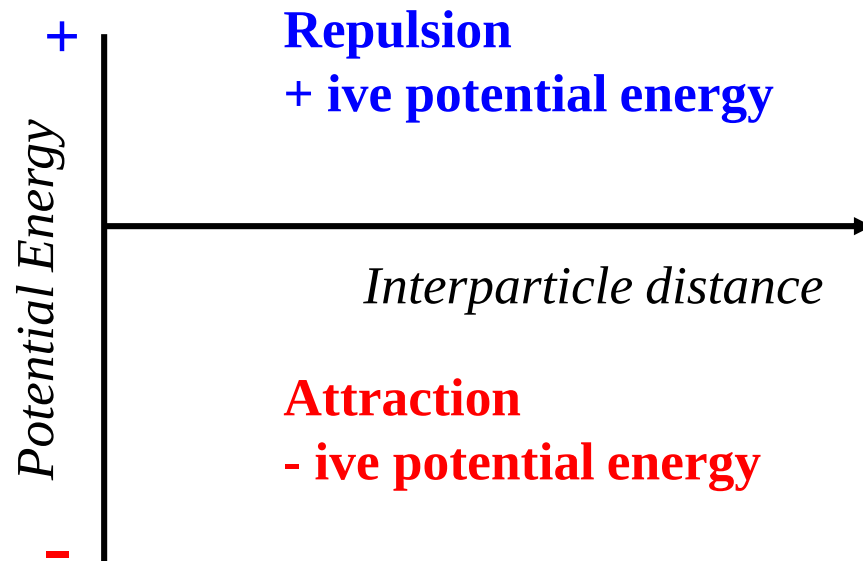
- *Several of these*
- *Magnitude dependent on distance between particles*
- *Attractive or repulsive*



As particles come close => repulsive forces in this case

Non-dilute Suspensions-Interparticle forces

Interparticle forces (attractive or repulsive) are often looked from the stand point of potential energy



of